Q3. IIR filters.

(a).

it has 3 pairs of poles : $S_1 = \Omega_c \cdot e^{j\frac{7}{12}\pi}$

$S_1^* = \Omega_c \cdot e^{-j\frac{7}{12}\pi}$

$S_2 = \Omega_c \cdot e^{j\frac{9}{12}\pi}$

$S_2^* = \Omega_c \cdot e^{-j\frac{9}{12}\pi}$

$S_3 = \Omega_c \cdot e^{j\frac{11}{12}\pi}$

$S_3^* = \Omega_c \cdot e^{-j\frac{11}{12}\pi}$

$\frac{\pi}{2} + \frac{2k+1}{N} \cdot \frac{\pi}{2}$

$\frac{2\pi}{12} = \frac{\pi}{6}$

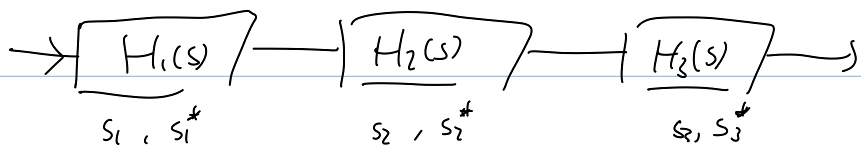
$\frac{\pi}{2} + \frac{1}{2} \cdot \frac{\pi}{6}$

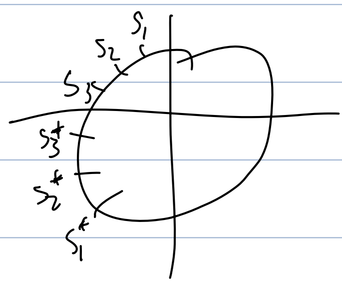
$\frac{8}{2}$

$\vdots$

it can be divided into 3 2-order system :

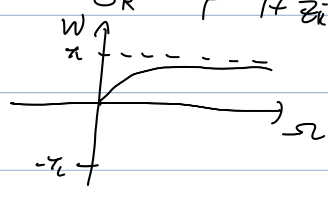
$$H(s) = \prod_{k=1}^{3} \frac{1}{(s-s_k)(s-s_k^*)}$$





(b) bilinear transform.

$$S_k = \frac{2}{T} \cdot \frac{1 - z_k^{-1}}{1 + z_k^{-1}} \Rightarrow z_k = \frac{1 + \frac{T}{2} S_k}{1 - \frac{T}{2} S_k} \Rightarrow H(z) \Rightarrow h(n)$$



$\Omega = \infty \rightarrow \omega = \pi$

$\downarrow$

Nyquist

* preserve stability (left plane →

* it maps $\Omega = \infty$ to $\omega = \pi$, so the Nyquist freq. experience the max. attenuation.

(c). Yes there will be.

$$H(z) = \frac{K}{\prod_k \left(\frac{2}{T} \cdot \frac{1-z^{-1}}{1+z^{-1}} - s_k\right)} = \frac{K(1+z^{-1})^k}{\prod_k \left(\frac{2}{T}(1-z^{-1}) - s_k(1+z^{-1})\right)}$$

$z = -1$, k. zeros.